Speed of Sound Problems

1. Explain why the speed of sound in wood may be between 4000 and 5300 m/s.

Substance	Speed of Sound (m/s)
Air at 0 ^{0} C	331
Aluminium	5100
Steel	4700-5200
Wood	4000-5300
Rubber	60
Seawater	1540
Freshwater	1410

2. Why is the speed of sound different for sea water and fresh water?

3. In calculating the depth of water under a ship, a person sends a sound wave to the bottom of the ocean. An echo is picked up 8.4 seconds later.

   a. How deep is the water?

   b. What would the depth be if the ship was on a freshwater lake, not the sea?

4. A ship travelling across the ocean is plotting the shape of the sea bed using SONAR. The ship sends a pulse every 10m of travel and collects the resultant echo.

Distance travelled (m)	Time for Echo (s)	Depth (m)
0	2.1	1617
10	3.4	
20	5.6	
30	6.2	
40	6.2	
50	5.2	
60	4.4	
70	3.4	
80	2.3	
90	5.8	

A ship travelling across the ocean is plotting the shape of the sea bed using SONAR. The ship sends a pulse every 10m of travel and collects the resultant echo.

100	4.6	

a. Complete depth column in the table. The first one has been done for you.

b. Plot the shape of the sea bed using the distance travelled and the depth.